

Week 2 Introduction

Mean, Variance and Standard Deviation

- ✓ **Video:** 2.1 – Introduction to Mathematical Concepts of Data Clustering
1 min
- ✓ **Practice Quiz:** Population vs Sample – Review Information
3 questions
- ✓ **Video:** 2.2 – Mean of One Dimensional Lists
2 min
- ✓ **Reading:** Population vs Sample, Bias
10 min
- ✓ **Practice Quiz:** Mean of One Dimensional Lists – Review Information
1 question
- ✓ **Video:** 2.3 – Variance and Standard Deviation
3 min
- ✓ **Reading:** Variability, Standard Deviation and Bias
10 min
- 📋 **Practice Quiz:** Variance and Standard Deviation – Review Information
1 question

Development Environment: Jupyter Notebooks

Computing Basic Statistics in Python

Week 2 Outro and Assessment



Variability, Standard Deviation

Variability

There is more variability if your data is distributed far from the mean, le

You can indeed have the same mean but high, or low, variability! As you and y are not the same, yet the red line representing their respective m

```

1  # two different arrays
2  x = np.array([2,3,5,6])
3  y = np.array([-8,-7,15,16])
4
5  # needed for the graph below
6  my_range = np.arange(4)
7
8  # their respective means
9  mean_x = np.mean(x)
10 mean_y = np.mean(y)
11
12 fig, axs = plt.subplots(1,2, sharey=True)
13 axs[0].stem(my_range, x, label='x values', bottom=True)
14 axs[1].stem(my_range, y, label='y values', bottom=True)
15 legend_x = axs[0].legend(loc='upper right')
16 legend_y = axs[1].legend(loc='lower right')
17 plt.show()
```

Remember this important idea: "the average of deviations is always zero" about why that is. If you look at the above two charts, that may help you *variability* of your data around the mean, then by adding all these numbers to the mean is, you could say, the middle of all these deviations, where they ca

We then use the square of the values to obtain more information: when cancellation of all deviations has been avoided. Now our deviations can be calculating the variance:

$$s^2 = \frac{\sum (x_i - \bar{x})^2}{n-1}.$$

When we are done with our calculation, and have computed the standard deviation, we are now taking the square root in our equation, after having squared ev

$$s = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n-1}}.$$